AI vs ML vs DL

You are my **Machine Learning Mentor, Data Scientist Interview Trainer, and Case Study Instructor**.
I am following a **60-day / 2-month Machine Learning roadmap** to become **job-ready for a Data Scientist role** And its second day teach me according to it
Cover topic such as theory mathematical intituition ,advantages ,Disadvantages , applications ,python code practice ,1project ,30 mcq questions and Potential interview questions

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🚀 Day 2 — AI vs Machine Learning vs Deep Learning

Today we will move from **"What is ML?"** to understanding exactly **where Machine Learning fits inside AI**, how Deep Learning fits inside ML, and how a Data Scientist actually uses these concepts.

Your Day 2 lesson is structured for a **Data Scientist job-ready path**:

1. Core theory
2. AI vs ML vs DL
3. Mathematical intuition
4. Traditional programming vs ML
5. Advantages & disadvantages
6. Real-world applications
7. Python coding demonstration
8. Mini project
9. 30 MCQs
10. Interview questions + answers
11. Day-2 interview/coding challenge

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1. The Big Picture

The easiest way to remember the relationship is:

```
Artificial Intelligence (AI)
│
├── Machine Learning (ML)
│   │
│   ├── Classical ML
│   │   ├── Linear Regression
│   │   ├── Logistic Regression
│   │   ├── Decision Trees
│   │   ├── Random Forest
│   │   ├── XGBoost
│   │   └── SVM
│   │
```

```
│       └── Deep Learning (DL)
│               ├── Neural Networks
│               ├── CNN
│               ├── RNN
│               ├── LSTM
│               └── Transformers
│
└── Other AI approaches
        ├── Rule-based systems
        ├── Search
        ├── Planning
        └── Expert systems
```

Key relationship

> **AI is the broadest concept. ML is a subset of AI. Deep Learning is a subset of ML.**

```
AI
 └── ML
      └── Deep Learning
```

But remember:

Not all AI is ML.

For example, a rule-based expert system can be AI without learning from data.

---

2. What Is Artificial Intelligence?

Artificial Intelligence is the broader field of creating systems capable of performing tasks that normally require some form of human intelligence.

Examples:

- reasoning
- decision making
- planning
- perception
- language understanding
- problem solving
- learning

Example

A chess-playing system needs to:

```
Understand board
       ↓
Evaluate positions
       ↓
Consider possible moves
       ↓
Choose move
```

This can involve AI techniques.

3. What Is Machine Learning?

Machine Learning focuses on systems that **learn patterns from data**.

Instead of manually writing every rule:

```
Rules + Data
     ↓
   Output
```

we provide examples:

```
Historical Data
      +
Known Outcomes
      ↓
  ML Algorithm
      ↓
   ML Model
      ↓
Prediction
```

Example: Loan Default

Suppose we have:

Income	Credit Score	Loan	Default
30,000	580	500,000	Yes
80,000	750	300,000	No
45,000	620	600,000	Yes
100,000	790	400,000	No

The ML algorithm tries to discover patterns.

For example:

```
Lower credit score
       +
Higher loan burden
       ↓
Higher probability of default
```

The model learns this relationship from historical data.

---

4. What Is Deep Learning?

Deep Learning is a specialized branch of Machine Learning based primarily on **multi-layer neural networks**.

Traditional ML often relies heavily on human-designed features.

Deep Learning can learn representations/features automatically from sufficiently large datasets.

Traditional ML

```
Raw Data
    ↓
Feature Engineering
    ↓
ML Algorithm
    ↓
Prediction
```

Deep Learning

```
Raw Data
    ↓
Neural Network
    ↓
Learned Features
    ↓
Prediction
```

5. Example: Image Recognition

Suppose you want to identify whether an image contains a cat.

Traditional ML

You might manually create features:

```
Ear shape
Eye size
Face shape
Fur texture
Whisker patterns
...
```

Then:

```
Features → Random Forest/SVM → Cat/Not Cat
```

Deep Learning

You can give images directly to a neural network:

```
Image
 ↓
Neural Network
 ↓
Edges
 ↓
Shapes
 ↓
Patterns
 ↓
Object representation
 ↓
Cat probability
```

The neural network learns useful representations from the images.

---

6. AI vs ML vs DL

Aspect	AI	ML	Deep Learning
Scope	Broadest	Subset of AI	Subset of ML
Main idea	Intelligent behavior	Learn from data	Learn representations using neural networks
Data requirement	Varies	Usually moderate	Often large
Feature engineering	May/may not	Often important	Often reduced
Compute	Varies	Usually moderate	Often high
Interpretability	Varies	Often better	Often harder
Examples	Expert systems, planning	Random Forest, XGBoost	CNN, LSTM, Transformers
Typical applications	Robotics, planning	Fraud, churn, credit risk	Vision, speech, LLMs

7. Mathematical Intuition

This is extremely important for Data Scientist interviews.

Think of an ML model as a mathematical function:

$$\hat{y} = f(X; \theta)$$

Where:

- X = input/features
- y = actual target
- $\hat{y}$ = prediction
- f = model
- θ = model parameters

Example

Suppose we're predicting house prices.

Features:

$$X = [Area, Bedrooms, Location]$$

Target:

$$y = Price$$

The model learns:

$$Price = f(Area, Bedrooms, Location)$$

8. Linear Regression Example

A simple linear regression model:

$$\hat{y} = \beta_0 + \beta_1 x$$

Suppose:

$$\hat{y} = 50000 + 3000x$$

where x = number of bedrooms.

For 3 bedrooms:

$$\hat{y} = 50000 + 3000(3)$$

$$= 59000$$

The model predicts:

₹59,000

The parameters are:

- $\beta_0 = 50000$
- $\beta_1 = 3000$

---

9. How Does ML Learn?

This is the core mathematical intuition.

Suppose actual value is:

$$y = 60,000$$

Model predicts:

$$\hat{y} = 59,000$$

Error:

$$Error = y - \hat{y}$$

$$= 60,000 - 59,000$$

$$= 1,000$$

But one error isn't enough.

We need a mathematical measure of overall model performance.

That's where **loss functions** come in.

---

10. Loss Function

For regression, one common loss is Mean Squared Error:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

The model tries to minimize this error.

Conceptually:

```
Bad parameters
      ↓
Large error
      ↓
Adjust parameters
      ↓
Smaller error
      ↓
Better parameters
```

This is the foundation of machine learning optimization.

---

11. Optimization Intuition

Imagine you're standing on a mountain and want to reach the lowest point.

```
         ●
        / \
       /   \
      /     \
     /       \
___/_________\___
       ↓
    Minimum
```

The algorithm takes steps toward lower loss.

One famous optimization algorithm is:

Gradient Descent

The simplified update equation is:

$$\theta_{new} = \theta_{old} - \alpha \nabla J(\theta)$$

where:

- θ = parameters
- α = learning rate
- $J(\theta)$ = loss function
- $\nabla J(\theta)$ = gradient

You don't need to master calculus today, but understand this:

> **The gradient tells the algorithm which direction increases the error, so we move in the opposite direction.**

---

12. Traditional Programming vs Machine Learning

Traditional Programming

```
Rules
  +
Data
  ↓
Program
  ↓
Output
```

Example:

```python
if age >= 18:
    status = "Adult"
else:
    status = "Minor"
```

The rule is explicitly written.

Machine Learning

```
Data
+
Examples/Labels
      ↓
Algorithm
      ↓
Model
      ↓
Prediction
```

The rules aren't explicitly written by the programmer.

The model learns patterns.

---

13. Advantages of Machine Learning

1. Automation

Can automate repetitive decisions.

Example:

```
Millions of transactions
        ↓
Fraud detection model
```

2. Pattern discovery

ML can identify relationships that are difficult to manually program.

---

3. Scalability

A trained model can process enormous amounts of data.

---

4. Personalization

Examples:

- Netflix recommendations
- Amazon recommendations
- Spotify recommendations

---

5. Prediction

Examples:

- customer churn
- loan default
- demand forecasting
- stock risk
- equipment failure

---

14. Disadvantages of Machine Learning

1. Requires quality data

Garbage in → garbage out.

---

2. Bias

If training data is biased:

```
Biased data
     ↓
Biased model
     ↓
Biased decisions
```

3. Overfitting

The model may memorize training data instead of learning general patterns.

---

4. Computational cost

Large models can require significant:

- CPU
- GPU
- memory
- storage

---

5. Interpretability

Some complex models are difficult to explain.

For example:

```
Neural Network
       ↓
Prediction
```

Why exactly did it make that prediction?

Sometimes difficult to answer.

---

15. Applications

Banking

Very relevant to your Data Scientist preparation.

Credit risk

```
Customer information
       ↓
ML model
       ↓
Probability of default
```

Fraud detection

```
Transaction
     ↓
Fraud model
     ↓
Fraud probability
```

Customer churn

```
Customer behavior
       ↓
ML model
       ↓
Churn probability
```

Healthcare

- disease prediction
- medical image analysis
- drug discovery
- patient risk prediction

E-commerce

- recommendations
- demand forecasting
- customer segmentation
- price optimization

Finance

- fraud detection
- credit scoring
- algorithmic trading
- risk modeling
- portfolio optimization

Manufacturing

- predictive maintenance
- defect detection
- quality control

---

16. Classical ML vs Deep Learning

Problem	Classical ML	Deep Learning
Tabular banking data	⭐⭐⭐⭐⭐	⭐⭐⭐
Credit risk	⭐⭐⭐⭐⭐	⭐⭐⭐
Customer churn	⭐⭐⭐⭐⭐	⭐⭐⭐
Image recognition	⭐⭐	⭐⭐⭐⭐⭐
Speech recognition	⭐	⭐⭐⭐⭐⭐

Problem	Classical ML	Deep Learning
NLP	⭐⭐⭐	⭐⭐⭐⭐⭐
Small dataset	⭐⭐⭐⭐⭐	⭐⭐
Huge dataset	⭐⭐⭐	⭐⭐⭐⭐⭐
Interpretability	Usually better	Usually harder

Important interview point

Don't assume:

> Deep Learning is always better than Machine Learning.

For **structured/tabular business data**, algorithms such as:

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM

can be excellent choices.

---

17. Python Demonstration

Let's create a very small classification problem.

We want to predict whether a customer will churn.

```python
import pandas as pd

data = {
    "age": [22, 25, 30, 35, 40, 45, 50, 55],
    "monthly_spend": [20, 25, 30, 40, 50, 60, 70, 80],
    "support_calls": [5, 4, 3, 2, 2, 1, 1, 0],
    "churn": [1, 1, 1, 0, 0, 0, 0, 0]
}

df = pd.DataFrame(data)

print(df)
```

Separate features and target

```python
X = df[["age", "monthly_spend", "support_calls"]]

y = df["churn"]
```

Remember:

```
X = Features
y = Target
```

18. Train/Test Split

```python
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)
```

Conceptually:

```
100% Data
    │
    ├── 80% Training
    │
    └── 20% Testing
```

Training data teaches the model.

Testing data evaluates generalization.

---

19. Train Logistic Regression

```python
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()

model.fit(X_train, y_train)
```

The model learns parameters from the training data.

---

20. Prediction

```python
predictions = model.predict(X_test)

print(predictions)
```

Probability:

```python
probabilities = model.predict_proba(X_test)

print(probabilities)
```

You might get something like:

```
[[0.20, 0.80]]
```

Meaning approximately:

```
Probability of no churn = 20%
Probability of churn    = 80%
```

21. Evaluate

```python
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, predictions)

print("Accuracy:", accuracy)
```

Important

Don't blindly use accuracy for every Data Science problem.

For example:

```
1000 customers
990 don't churn
10 churn
```

A model predicting:

```
Everyone = No Churn
```

gets:

$$990/1000 = 99\%$$

accuracy.

Yet the model is useless for finding churners.

This leads us to metrics such as:

- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC

You'll study these in detail later.

🧪 DAY 2 MINI PROJECT

Customer Churn Prediction

Business Problem

A telecom company wants to identify customers who are likely to leave.

The company wants to contact high-risk customers with retention offers.

Dataset

Create:

```python
data = {
    "age": [22,25,30,35,40,45,50,55,28,33,48,52],
    "monthly_spend": [20,25,30,40,50,60,70,80,35,45,65,75],
    "support_calls": [5,4,3,2,2,1,1,0,4,3,1,0],
    "tenure_months": [2,4,6,18,24,36,48,60,5,12,40,55],
    "churn": [1,1,1,0,0,0,0,0,1,1,0,0]
}

df = pd.DataFrame(data)
```

Your tasks

Task 1 — Explore

```python
print(df.head())
print(df.info())
print(df.describe())
```

Task 2 — Define X and y

```python
X = df.drop("churn", axis=1)
y = df["churn"]
```

Task 3 — Split

```python
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)
```

Task 4 — Train

```python
model = LogisticRegression()

model.fit(X_train, y_train)
```

Task 5 — Predict

```python
y_pred = model.predict(X_test)
```

Task 6 — Evaluate

```python
from sklearn.metrics import accuracy_score, classification_report

print("Accuracy:", accuracy_score(y_test, y_pred))

print(classification_report(y_test, y_pred))
```

Task 7 — Business interpretation

Answer:

1. Which customers are most likely to churn?
2. Which features appear related to churn?
3. Should the company target everyone?
4. What would happen if the company missed a high-risk churn customer?
5. Which metric might be more important: precision or recall?

This last part is **Data Scientist thinking**.

🧠 22. 30 MCQs — Day 2

Q1. AI is:

A. A subset of ML
B. A subset of DL
C. The broader field containing ML
D. Only neural networks

Answer: C

Q2. Machine Learning is generally considered:

A. A subset of AI
B. A subset of databases
C. A programming language
D. A hardware technology

Answer: A

Q3. Deep Learning is:

A. Outside AI
B. A subset of ML
C. A database technique
D. A visualization library

Answer: B

Q4. Which is an example of Deep Learning?

A. Linear Regression
B. Decision Tree
C. CNN
D. SQL

Answer: C

Q5. Which is usually best suited to image recognition?

A. Deep Learning
B. Excel formulas
C. SQL joins
D. Linear Regression only

Answer: A

Q6. In supervised learning, the training data contains:

A. Only features
B. Features and target labels
C. Only labels
D. No data

Answer: B

Q7. X usually represents:

A. Target
B. Features
C. Loss
D. Prediction error

Answer: B

Q8. y usually represents:

A. Target
B. Feature
C. Learning rate
D. Model parameter

Answer: A

Q9. A model prediction is commonly represented as:

A. x
B. y
C. $\hat{y}$
D. θ only

Answer: C

Q10. What does a loss function measure?

A. Dataset size
B. Model error
C. Number of features only
D. CPU speed

Answer: B

Q11. MSE stands for:

A. Machine System Error
B. Mean Squared Error
C. Maximum Statistical Estimate
D. Model Selection Evaluation

Answer: B

Q12. Gradient Descent is used primarily for:

A. Visualization
B. Optimization

C. Data collection
D. Database management

Answer: B

Q13. The learning rate controls:

A. Dataset size
B. Step size during optimization
C. Number of columns
D. Number of labels

Answer: B

Q14. What happens if the learning rate is extremely large?

A. It can overshoot the minimum
B. Training always becomes perfect
C. Dataset becomes larger
D. Features disappear

Answer: A

Q15. Which is a classical ML algorithm?

A. Random Forest
B. CNN
C. Transformer
D. LSTM

Answer: A

Q16. Which is a Deep Learning architecture?

A. Random Forest
B. Decision Tree
C. CNN
D. K-Means

Answer: C

Q17. Deep Learning generally benefits from:

A. Large amounts of data
B. No data
C. Only two observations
D. Manual rules only

Answer: A

Q18. One advantage of traditional ML over DL can be:

A. Always more accurate
B. Better suitability for small tabular datasets
C. Requires more GPUs
D. Cannot work with structured data

Answer: B

Q19. Overfitting means:

A. Model performs well on unseen data
B. Model memorizes training patterns too closely
C. Model has no parameters
D. Dataset contains no labels

Answer: B

Q20. Why do we use test data?

A. To train the model
B. To evaluate generalization
C. To increase training data
D. To remove features

Answer: B

Q21. Which Python library is commonly used for classical ML?

A. scikit-learn
B. NumPy only
C. Flask
D. BeautifulSoup

Answer: A

Q22. Which function trains a scikit-learn model?

A. `.predict()`
B. `.fit()`
C. `.score_data()`
D. `.learn()`

Answer: B

Q23. Which function generates predictions?

A. `.fit()`
B. `.predict()`
C. `.train()`
D. `.compile()`

Answer: B

Q24. Which is a classification algorithm?

A. Logistic Regression
B. Linear Regression only
C. PCA only
D. StandardScaler

Answer: A

Q25. Which problem is classification?

A. Predict house price
B. Predict customer churn yes/no
C. Predict temperature
D. Predict salary amount

Answer: B

Q26. Which problem is regression?

A. Spam/not spam
B. Fraud/not fraud
C. House price prediction
D. Churn/not churn

Answer: C

Q27. Which statement is correct?

A. Every AI system is Deep Learning
B. Every ML system is Deep Learning
C. Deep Learning is a subset of ML
D. ML and AI are identical

Answer: C

Q28. Why can accuracy be misleading?

A. It always equals zero
B. It can hide poor performance on minority classes
C. It cannot be calculated
D. It only works with regression

Answer: B

Q29. A model that performs extremely well on training data but poorly on test data likely suffers from:

A. Underfitting
B. Overfitting
C. Normalization
D. Sampling

Answer: B

Q30. Which statement best describes ML?

A. Humans explicitly program every decision rule
B. Models learn patterns from data and generalize to new data
C. ML requires no data
D. ML only means neural networks

Answer: B

🎯 23. Data Scientist Interview Questions

Q1. What is the difference between AI, ML and Deep Learning?

Answer

AI is the broad field of creating systems capable of intelligent behavior.

ML is a subset of AI where algorithms learn patterns from data.

Deep Learning is a subset of ML using multi-layer neural networks to learn complex representations.

```
AI
 └── ML
      └── DL
```

Q2. Is every AI system a Machine Learning system?

No.

AI can also use:

- rules
- search
- planning
- expert systems
- symbolic reasoning

Therefore:

> AI is broader than ML.

Q3. When would you choose traditional ML instead of Deep Learning?

A strong interview answer:

> I would generally consider traditional ML when the dataset is relatively small or medium-sized, especially for structured/tabular business data, when training resources are limited, or when interpretability is important. Algorithms such as logistic regression, random forests, and gradient-boosted trees can perform extremely well in these situations.

Q4. Why is Deep Learning powerful?

Because neural networks can learn increasingly complex representations through multiple layers.

For an image:

```
Pixels
 ↓
Edges
 ↓
Shapes
 ↓
Objects
```

```
 ↓
Classification
```

This reduces the need for manually designing every feature.

Q5. What is a feature?

A feature is an input variable used by the model.

Example:

```
Customer age
Monthly income
Loan amount
Credit score
```

These can be features.

Q6. What is a target?

The variable the model tries to predict.

Example:

```
Features:
age
income
credit_score

Target:
loan_default
```

Q7. Why split data into training and testing?

To determine whether the model generalizes to **unseen data**.

If we evaluate only on training data, we may get an overly optimistic estimate of performance.

Q8. What is overfitting?

Overfitting occurs when the model learns the training data too closely, including noise or idiosyncratic patterns, and therefore performs poorly on unseen data.

Q9. What is underfitting?

Underfitting occurs when the model is too simple to capture the important patterns in the data.

```
Underfitting → High bias
Good model   → Good generalization
Overfitting  → High variance
```

You'll study this deeply later.

Q10. What is generalization?

Generalization means the model performs well on **new, unseen observations**, not merely on its training examples.

This is one of the most important concepts in ML.

💼 24. Case Study — Banking Credit Risk

Since banking is an important Data Science domain, let's connect today's lesson to a real business scenario.

Problem

A bank wants to predict:

> "Will this applicant default on the loan?"

Data:

```
Age
Income
Credit Score
Employment Length
Loan Amount
Debt-to-Income Ratio
Past Defaults
```

Target:

```
Default = 1
No Default = 0
```

ML pipeline

```
Raw Banking Data
       ↓
Data Cleaning
       ↓
EDA
       ↓
Feature Engineering
       ↓
Train/Test Split
       ↓
ML Model
       ↓
Probability of Default
       ↓
Risk Decision
```

For example:

```
Customer A

PD = 0.03
```

means estimated probability of default is approximately **3%**.

Another customer:

```
Customer B

PD = 0.72
```

means estimated probability is approximately **72%**.

The bank could then use this probability as an input into its broader credit decision process.

Important Data Scientist point

The Data Scientist isn't simply:

> "Build a model."

The actual job is:

```
Business problem
      ↓
Data
      ↓
Statistical understanding
      ↓
Feature engineering
      ↓
Model
      ↓
Evaluation
      ↓
Business decision
      ↓
Monitoring
```

🧠 25. What You Should Remember From Day 2

If you remember only these **10 points**, you've succeeded today:

1.

AI is broader than ML.

2.

ML is a subset of AI.

3.

Deep Learning is a subset of ML.

4.

Traditional ML often works extremely well on **structured/tabular data**.

5.

Deep Learning is particularly powerful for complex unstructured data such as:

- images
- audio
- text
- video

6.

An ML model can be viewed mathematically as:

$$\hat{y} = f(X;\theta)$$

7.

The model learns parameters by minimizing a loss function.

8.

Gradient Descent is an optimization technique.

9.

Training performance isn't enough; we care about **generalization**.

10.

The ultimate goal of a Data Scientist isn't simply to build a model—it is to **solve a business problem using data**.

📝 Day 2 Assignment

Before moving to Day 3, complete these:

Theory

Explain in your own words:

1. AI vs ML vs DL
2. Traditional programming vs ML
3. Features vs target
4. Training vs testing
5. Overfitting vs underfitting

6. Loss function
7. Gradient Descent
8. Generalization

Coding

Build the **Customer Churn Prediction** project above and try to answer:

```
1. What is X?
2. What is y?
3. Why do we split the data?
4. What does .fit() do?
5. What does .predict() do?
6. Why might accuracy be misleading?
7. Which customers should the business target?
```

Interview challenge

Try answering this without looking above:

> **"A bank gives you 100,000 historical loan records. How would you build a machine learning system to predict loan default?"**

A strong Data Scientist should be able to explain the **complete pipeline**, not just name an algorithm.

Day 2 success criterion: You should now be able to explain **AI** → **ML** → **DL**, mathematically describe an ML model at a high level, write a basic scikit-learn model, and connect ML to a real banking/business problem.